

Your Name:

ID #:

### Worksheet: Inverse Trig Functions and their Domains and Ranges

1. Find the inverse of each function and list the domain and range of  $f^{-1}(x)$

(a)  $f(x) = 5 \sin(x) - 3$  for  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$$y = 5 \sin(x) - 3$$

$$\implies y + 3 = 5 \sin(x)$$

$$\implies \sin(x) = \frac{y+3}{5}$$

$$\implies x = \sin^{-1}\left(\frac{y+3}{5}\right)$$

$$\implies f^{-1}(x) = \sin^{-1}\left(\frac{x+3}{5}\right)$$

Find domain of  $f^{-1}(x)$  by finding the range of  $f(x)$ :

$$-1 \leq \sin x \leq 1$$

$$\implies -5 \leq 5 \sin(x) - 3 \leq 2$$

$$\implies -8 \leq y \leq 2$$

$$\text{Domain of } f^{-1}(x) : [-8, 2]$$

$$f^{-1}(x) = \sin^{-1}\left(\frac{x+3}{5}\right)$$

$$\text{Domain of } f^{-1}(x) : [-8, 2]$$

$$\text{Range of } f^{-1}(x) : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

(b)  $f(\theta) = \frac{1}{2} \tan(\theta)$  for  $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$$y = \frac{1}{2} \tan(\theta)$$

$$\implies 2y = \tan(\theta)$$

$$\implies \theta = \tan^{-1}(2y)$$

$$\implies f^{-1}(\theta) = \tan^{-1}(2\theta)$$

Find domain of  $f^{-1}(\theta)$  by finding the range of  $f(\theta)$ :

$$-\infty < \tan \theta < \infty$$

$$\implies -\infty < \frac{1}{2} \tan \theta < \infty$$

$$\implies -\infty < y < \infty$$

$$\text{Domain of } f^{-1}(\theta) : (-\infty, \infty)$$

$$f^{-1}(\theta) = \tan^{-1}(2\theta)$$

$$\text{Domain of } f^{-1}(\theta) : (-\infty, \infty)$$

$$\text{Range of } f^{-1}(\theta) : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

(c)  $f(x) = 4 - \cos(2x)$  for  $0 \leq x \leq \frac{\pi}{2}$

$$y = 4 - \cos(2x)$$

$$\implies y - 4 = -\cos(2x)$$

$$\implies \cos(2x) = 4 - y$$

$$\implies 2x = \cos^{-1}(4 - y)$$

$$\implies f^{-1}(x) = \frac{1}{2} \cos^{-1}(4 - x)$$

Find domain of  $f^{-1}(x)$  by finding the range of  $f(x)$ . For  $x \in [0, \frac{\pi}{2}]$ :

At  $x = 0$ :  $f(0) = 4 - \cos(0) = 4 - 1 = 3$

At  $x = \frac{\pi}{2}$ :  $f(\frac{\pi}{2}) = 4 - \cos(\pi) = 4 - (-1) = 5$

$$f^{-1}(x) = \frac{1}{2} \cos^{-1}(4 - x)$$

Domain of  $f^{-1}(x)$  :  $[3, 5]$

Range of  $f^{-1}(x)$  :  $[0, \frac{\pi}{2}]$

(d)  $f(\theta) = 4 - \sin(\theta - \pi) + 1$  for  $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$

$$f(\theta) = 5 - \sin(\theta - \pi), \theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$$

$$y = 5 - \sin(x - \pi)$$

$$\implies \sin(x - \pi) = 5 - y$$

$$\implies x - \pi = \sin^{-1}(5 - y) \text{ (the restriction of } \theta \text{ validate this step)}$$

$$\implies x = \sin^{-1}(5 - y) + \pi$$

$$\implies f^{-1}(\theta) = \sin^{-1}(5 - \theta) + \pi$$

Find domain of  $f^{-1}(\theta)$  by finding the range of  $f(\theta)$  :

$$\theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$$

$$\theta - \pi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\implies -1 \leq \sin(\theta - \pi) \leq 1$$

$$\implies 4 \leq 5 - \sin(\theta - \pi) \leq 6$$

$$\implies 4 \leq f(\theta) \leq 6$$

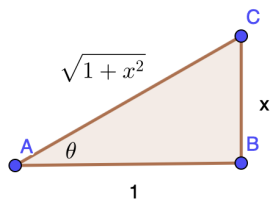
$$f^{-1}(\theta) = \sin^{-1}(5 - \theta) + \pi$$

Domain of  $f^{-1}(\theta)$  :  $[4, 6]$

Range of  $f^{-1}(\theta)$  :  $[\frac{\pi}{2}, \frac{3\pi}{2}]$

2. Write each trig expression as an algebraic expression and find its domain.

(a)  $\sin \tan^{-1}(x)$



Let  $\theta = \tan^{-1}(x)$ . Then,  $\tan(\theta) = x$ .

For a triangle, let:

- Opposite side =  $x$
- Adjacent side =  $1$
- Hypotenuse =  $\sqrt{1+x^2}$

$$\Rightarrow \sin(\tan^{-1}(x)) = \frac{x}{\sqrt{1+x^2}}$$

Domain of  $\tan^{-1}(x) : (-\infty, \infty)$

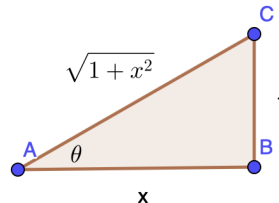
So,  $\boxed{\sin(\tan^{-1}(x)) = \frac{x}{\sqrt{1+x^2}}}$  and  $\boxed{\text{domain: } (-\infty, \infty)}$

(b)  $\csc \cot^{-1}(x)$

Let  $\theta = \cot^{-1}(x)$ . Then,  $\cot(\theta) = x$ .

For a triangle, let:

- Adjacent side =  $x$
- Opposite side =  $1$
- Hypotenuse =  $\sqrt{1+x^2}$



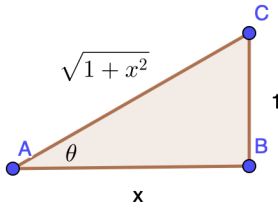
$$\Rightarrow \csc(\theta) = \frac{\text{hypotenuse}}{\text{opposite}} = \frac{\sqrt{1+x^2}}{1} = \sqrt{1+x^2}$$

$$\Rightarrow \csc(\cot^{-1}(x)) = \sqrt{1+x^2}$$

Domain of  $\cot^{-1}(x) : (-\infty, \infty)$

So,  $\boxed{\csc(\cot^{-1}(x)) = \sqrt{1+x^2}}$  and  $\boxed{\text{domain: } (-\infty, \infty)}$

(c)  $\sec \cot^{-1}(x)$



Let  $\theta = \cot^{-1}(x)$ . Then,  $\cot(\theta) = x$ .  
For a triangle, let:

- Adjacent side =  $x$
- Opposite side =  $1$
- Hypotenuse =  $\sqrt{1+x^2}$

$$\begin{aligned} \Rightarrow \sec(\theta) &= \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{\sqrt{1+x^2}}{x} \\ \Rightarrow \sec(\cot^{-1}(x)) &= \frac{\sqrt{1+x^2}}{x} \end{aligned}$$

Domain of  $\cot^{-1}(x) : (-\infty, \infty)$

However, when  $x=0$ ,  $\sec \cot^{-1} 0$  is undefined,

which corresponds to  $\frac{\sqrt{1+x^2}}{x}$  is undefined at  $x = 0$ .

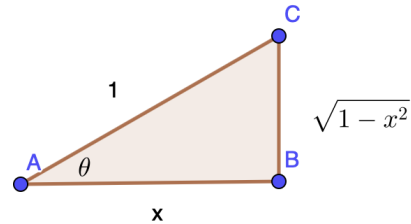
So,  $\boxed{\sec(\cot^{-1}(x)) = \frac{\sqrt{1+x^2}}{x}}$  and  $\boxed{\text{domain: } (-\infty, 0) \cup (0, \infty)}$

(d)  $\sec \cos^{-1}(x)$

Let  $\theta = \cos^{-1}(x)$ . Then,  $\cos(\theta) = x$ .

For a triangle, let:

- Adjacent side =  $x$
- Hypotenuse =  $1$
- Opposite side =  $\sqrt{1-x^2}$



$$\begin{aligned} \Rightarrow \sec(\theta) &= \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{1}{x} \\ \Rightarrow \sec(\cos^{-1}(x)) &= \frac{1}{x} \end{aligned}$$

Domain of  $\cos^{-1}(x) : [-1, 1]$

However, when  $x=0$ ,  $\sec \cos^{-1} 0$  is undefined,

which corresponds to  $\frac{1}{x}$  is undefined at  $x = 0$ .

So,  $\boxed{\sec(\cos^{-1}(x)) = \frac{1}{x}}$  and  $\boxed{\text{domain: } [-1, 0) \cup (0, 1]}$